Monte Carlo Simulations of Electron in Wire

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April 2026

0.1 Introduction

The objective of this project is to use a Monte Carlo Simulation to independently verify the Drude model's approximation for the drift velocity of an electron in a copper wire with an applied electric field, showing the clear application of randomness to small-scale particle interaction via probabilistic distributions. The theory of electron drift is grounded in modeling the electron as a classical moving particle with very high thermal velocity–and a relatively small velocity due to the electric field within the wire–scattering many times off the copper ions of the wire, given by:

$$v_{dr} = \frac{\lambda_f e E}{2 m_e v_{th}}$$

Where λ_f is the mean free path, or the average distance between collisions.

A Monte Carlo simulation is a well-suited approach for this objective because drift emerges statistically over the many scattering events. We will consider two main probabilistic parameters: the scattering angles off of particles θ and ϕ, and the mean free time τ, related to the mean free path via $\tau = \frac{\lambda_f}{v_{th}}$. By running many simulations of electron trajectories using these stochastic processes, we can estimate the average drift and see if the $\frac{1}{2}$ factor in the Drude drift velocity formula emerges purely from the statistical properties of random scattering and exponentially distributed free-flight times.

0.2 Framework

Within a copper wire, electrons move within the atomic lattice as a result of their thermal energy. These electrons collide regularly with phonons, which are vibrations of the lattice that act as quasiparticles. In the Drude model, these collisions are classical and perfectly elastic (like two billiard balls). This is our first source of randomness. There is no way to predict the true trajectory of the electron in relation to the phonons around it, and so we must make assumptions about how these collisions change the electrons velocity. Because the collisions are said to be elastic, the magnitude of the velocity is preserved under each collision, and only the direction is changed. Thus, we treat a collision as a rotation on a sphere, where every rotation is equally likely. To apply the rotation, we must sample random variables θ and ϕ, where θ is the polar angle and ϕ the azimuthal angle. Our equal probability suggests the use of uniform distributions, but closer scrutiny is needed. While choosing ϕ to be uniform on $(0, 2\pi)$ equally samples all points on the circumference, the same is not true for a uniform θ on $(0, \pi)$, as this over-represents the poles and under-samples the equatorial regions. Hence, for our collision parameters, we have:

$$\phi \sim U(0, 2\pi), \quad cos(\theta) \sim U(-1, 1)$$

In the presence of an electric field within the wire, the electron experiences an electric force, causing acceleration in the opposite direction of the field during the time between collisions, given by:

$$F_e = -eE$$

A quick application of Newton's second law shows:

$$m\dot{v}_{dr} = -eE \iff v_{dr}(t) = -\frac{eE}{m_e}t$$

Here we see the probabilistic parameter for the free-flight of our electron, in the variable t. The $v_{dr}(t)$ acts only between collisions, and hence, to obtain the drift velocity, we must consider the time between collisions. The most illuminating way to do this is to think about the distribution of collisions themselves, which are independent, unlikely to occur simultaneously, and should occur at an average rate within intervals of time. These are the clear characteristics of a Poisson process, so the time between them should be exponential. Hence, we will take our time between collisions to be exponential with an average τ.

$$t \sim Exp(\tau)$$

0.3 Expected Values and Theoretical Drift

With the distributions of our random variables established, we may investigate their contribution to the velocity over many collisions. Because ϕ is evenly distributed on $(0, 2\pi)$, its contribution to v_x averages to zero. This shows that ϕ affects only the rotational symmetry, while θ determines the distribution of velocity along the drift component. Hence, we investigate $\mathbb{E}[cos(\theta)]$ and see:

$$\mathbb{E}[cos(\theta)] = \frac{-1+1}{2} = 0$$

Therefore, the average velocity after a collision has no preferred direction. We will refer to this velocity directly after a collision as $v(0)$.

Now, to find the drift velocity we must consider the average velocity between any two collisions:

$$v_x^{mean} = \frac{1}{t}\int_0^t (v_x(0) + at')dt' = v_x(0) + \frac{1}{2}at$$

From this we can find the drift velocity:

$$v_{dr} = \mathbb{E}[v_x^{mean}] = \mathbb{E}[v_x(0) + \frac{1}{2}at] = \mathbb{E}[v_x(0)] + \frac{1}{2}a\mathbb{E}[t] = \frac{1}{2}a\tau$$

0.4 Simulation Methodology

To test whether the probabilistic picture of the Drude model produces the theoretical drift velocity, we simulate many electron trajectories, each consisting of 10^5

free-flight and rotation cycles. This is because the size of v_{th} is much greater than that of the theoretical v_{dr}, such that even for very strong electric fields it would take $> 10^8$ iterations to accurately differentiate the result from the noise. Running many shorter simulations allows us to find the distribution of v_{dr} and recover its experimental value by taking the mean.

Each simulation begins with an electron moving at speed v_{th} in a random direction, initialized by sampling a 3-dimensional gaussian vector and normalizing it. During each free-flight, v_x is updated using $v_x(t) = v_x(0) + at$, where t is sampled from an exponential distribution with parameter τ. As the Drude model corresponds to the per-flight average velocity, the simulation saves the average over the flight, $v_x = v_x(0) + \frac{1}{2}at$. For each collision, the direction of v is randomized while preserving its length. The angles are sampled according to:

$$\phi \sim U(0, 2\pi), \quad cos(\theta) \sim U(-1, 1)$$

The vector is rotated using an orthonormal basis constructed from the pre-collision direction. For each trajectory, the simulation find the drift velocity by taking the average of the per-flight mean velocities:

$$v_{dr} = \frac{1}{N}\sum_{n=1}^{N} v_{x,n}^{mean}$$

This quantity converges to $\frac{1}{2}a\tau$ as N becomes very large.

0.5 Results

The simulations used the copper-specific values of λ_f and τ, with the magnitude of the electric field being $E = 10^6$, pointing in the negative x direction.

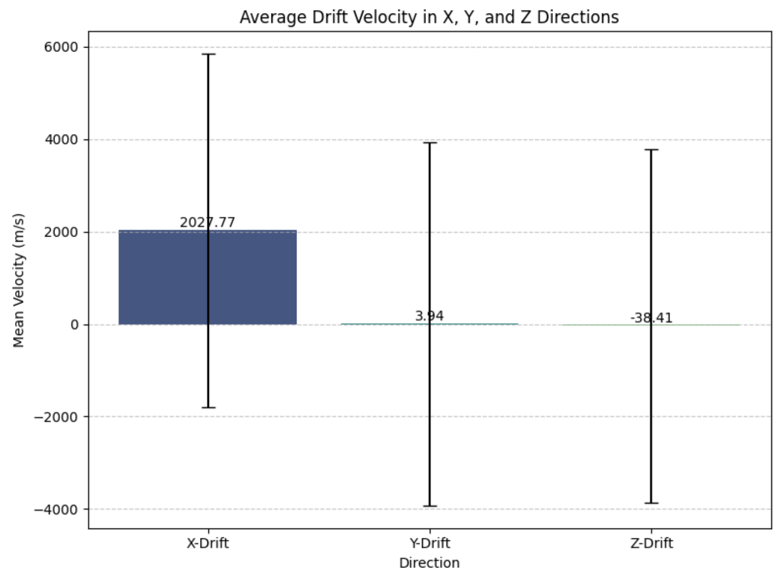

Figure 1: Average Drift in X, Y, and Z

The bar chart represents the mean velocity components against all simulations. As expected, the average y and z components are practically zero (3.04 m/s and -38.41 m/s, respectively), which is negligible compared to the magnitude of thermal velocity $(1.57 \cdot 10^6)$. From this, we can conclude that the rotations from our sampled $cos(\theta)$ and ϕ do not introduce a bias towards any direction.

For the x component, however, Figure 1 shows a distinct positive positive value of 2027.77 m/s. This represents our v_{dr} over all simulations, as our E for the simulation pointed in the negative x direction, so we would expect any contributions from the electric force to be in the positive x direction. The large error bars show that any one simulation still contains large amounts of noise from v_{th}, but the mean is consistent with out model.

Parameter	Value (m/s)
Theoretical Drift Velocity ($v_{dr} = \frac{1}{2}a\tau$)	2184.61
Simulated Mean X-Velocity ($v_{dr,sim}$)	2027.77
Percent Error	**7.18%**

Table 1: Theoretical Prediction and Monte Carlo Results

As shown in the table above, the result for v_{dr} from our simulation matches very closely with the theoretical prediction acquired using the Drude model. While our error of 7.18% is significant, it is expected for relatively short runs (compared to the $\sim 10^8$ required for convergence), and supports that as N increases, the simulated value would converge to the Drude model's prediction, with the factor of $\frac{1}{2}$.

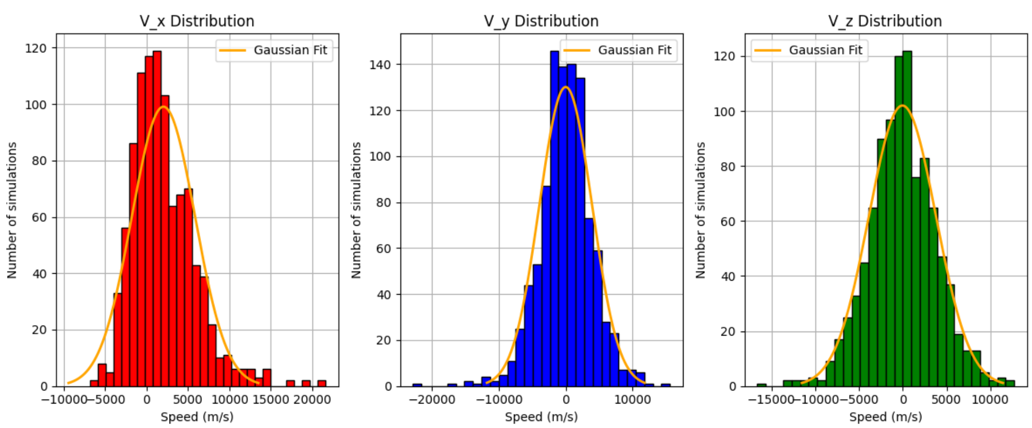

Figure 2: Distributions of Mean Velocities in X, Y, and Z

These histograms illustrate the statistical convergence of the simulation, where each velocity competent follows a gaussian distribution about it's mean (the estimated distributions are shown in orange). While the V_y and V_z distributions are centered around zero, the V_x distribution shows a clear shift that correlates with the collective drift of the many trajectories. This transition from individual stochastic scattering events to a predictable macroscopic drift confirms that the $\frac{1}{2}$ factor in the Drude model is a direct consequence of the underlying Poisson process of collisions. Ultimately, our Monte Carlo approach successfully validates the classical theory of conduction, demonstrating that even within a high-variance system dominated by thermal noise, statistical averaging can uncover profound (yet relatively tiny) behavior.

Bibliography

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